



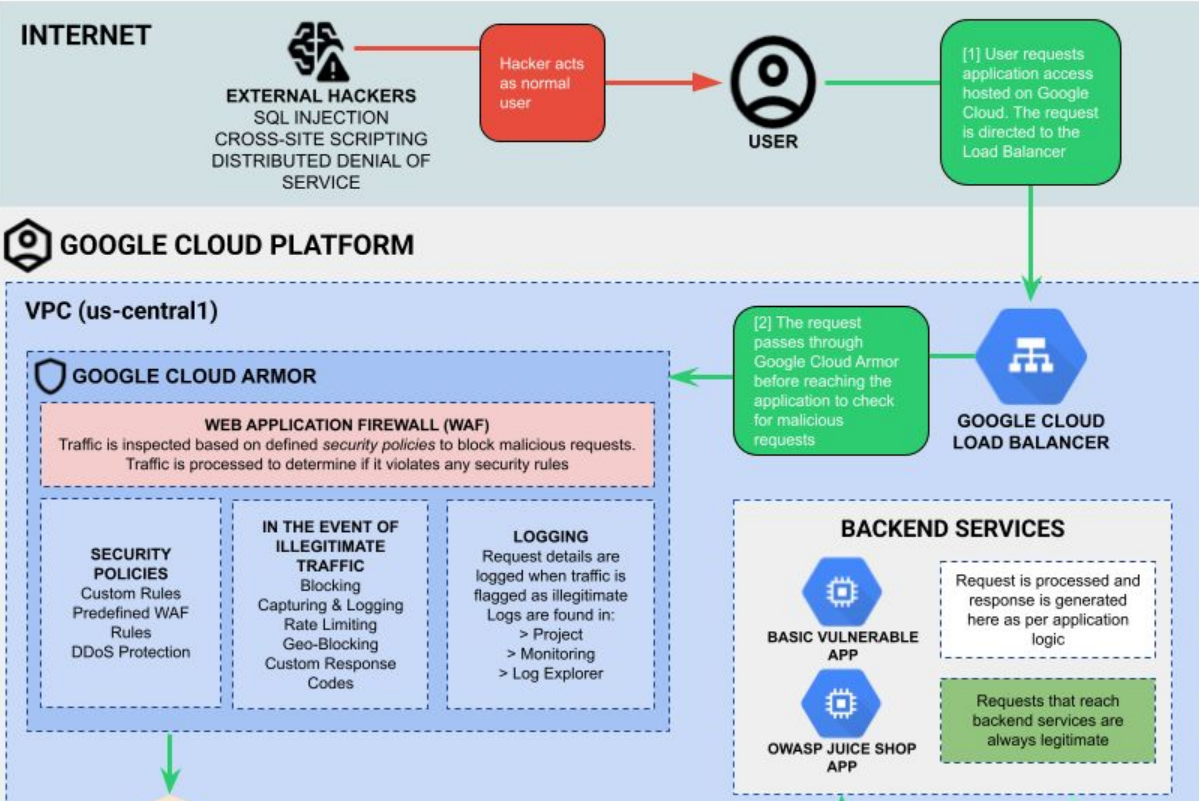
Google Cloud Armor WAF

Timothy Loudon, Melvin Viado, Daniel Lindo

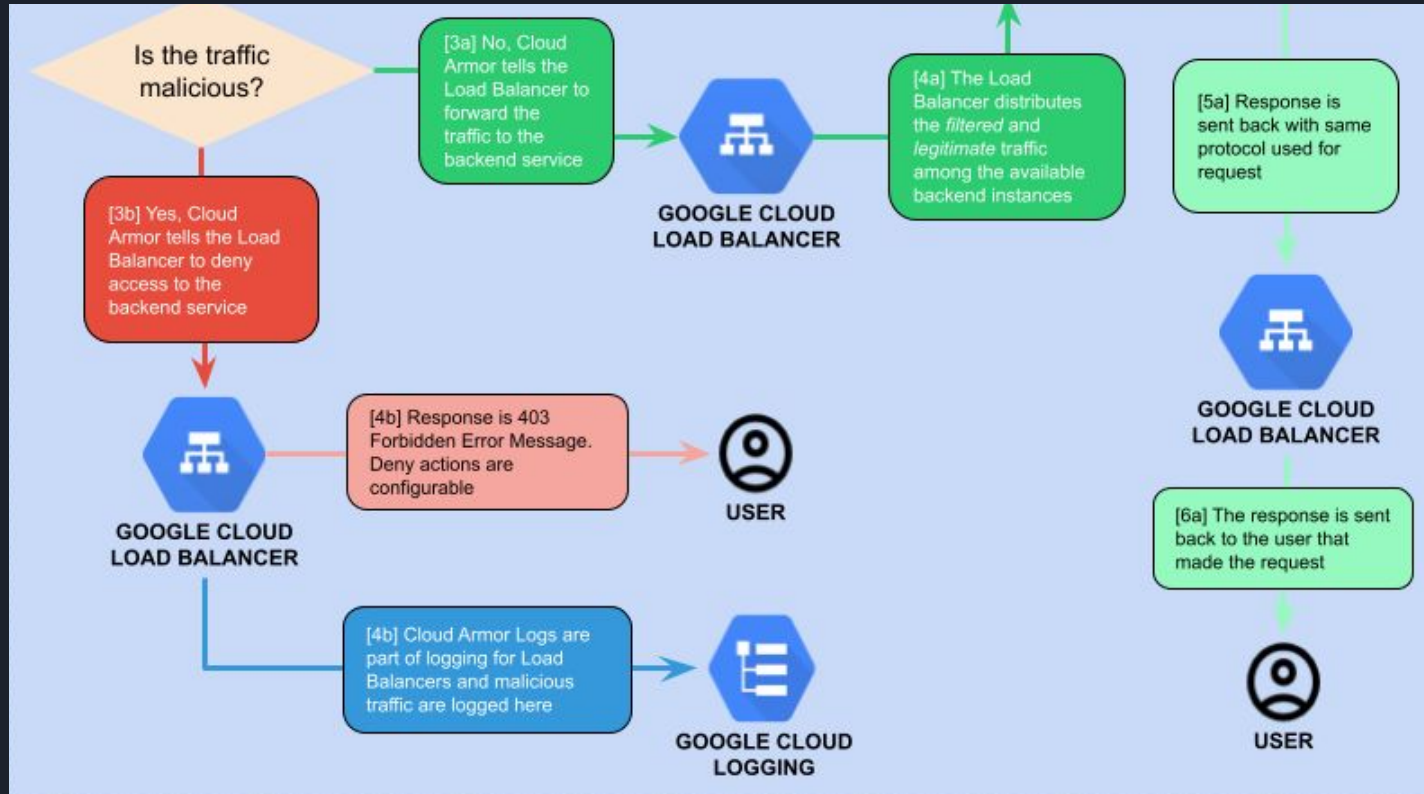
Traffic Flows

INFT1204 Assignment 3 Design Phase - Design of Network Infrastructure and Network Flow - Group 6 (Melvin Viado, Tim Loudon, Daniel Lindo)

■ Legitimate Request Traffic ■ Legitimate Response Traffic ■ Malicious Request Traffic ■ Malicious Response Traffic ■ Logging Data



Traffic Flows Cont.



OpenSSL & the Configuration of the VM Hosting OWASP Juice Shop and Docker



SSL Certificate Breakdown

- The Google Cloud Armor load balancer requires the SSL certificate to handle HTTPS traffic

- A Google certified SSL certificate requires one having a domain, which we do not have

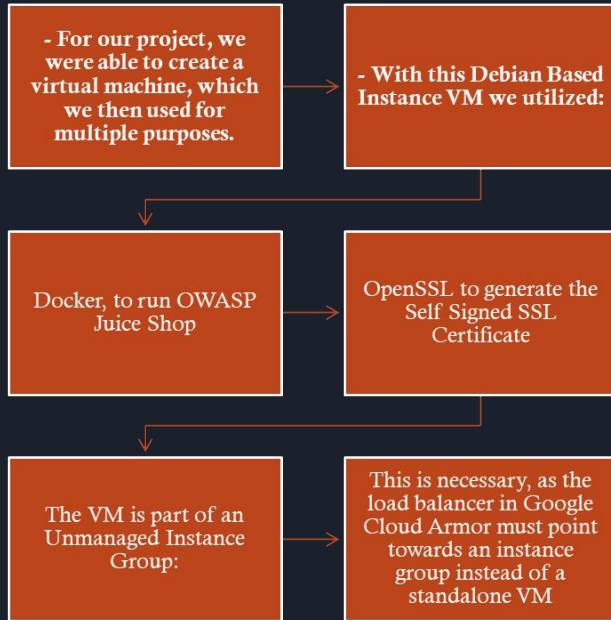
- Our certificate is self signed, hence the warning that appears in browsers, as the certificate is not signed by a publicly recognized certificate authority

- Once traffic reaches the load balancer, it is decrypted once inspected and sent to the web application.

- After reaching the web app, it is then sent back to the end user, after being encrypted by the load balancer

- If the load balancer finds the traffic to be suspicious or from an attacker, the packets are dropped and will not reach the web application

How OpenSSL Ties in and Docker and the Load Balancer within the VM



- Pros:

Generating the SSL certificate allows for the data transmitted between the end user and the load balancer to be encrypted until it reaches the Juice Shop web application

We also had a seamless integration with running juice shop within Docker on our virtual machine hosting the Juice Shop

- Cons:

Our SSL certificate to enable encryption (HTTPS) is self signed, hence the warning received within the browser when attempting to reach the Juice Shop web app

The Components - Virtual Machine

VM instances

Filter Enter property name or value

☐ Status	Name ↑	Zone
☐	test-vm	us-central1-a

```
Tim's Personal Email @test-vm:~$ ls
server.crt  server.csr  server.key
```

Our html page and OWASP docker image also lives here

Instance groups

Create instance group

Refresh

Delete

Instance groups are collections of VM instances that use load balancing and automated services, like auto-scaling and auto-healing. [Learn more](#)

Filter Enter property name or value

☐ Status	Name ↑	Instances	Template	Group type
☐	special-topics-instance-group	1	-	Unmanaged

The Components - Load Balancer

Load balancers

Backends

Frontends

Service LB policies

Filter

Enter property name or value

☐	Name	Load balancer type	Access type	Protocols
☐	special-topics-load-balancer	Application	External	HTTPS

Load balancers	Backends	Frontends	Service I
Filter Enter property name or value			
☐	Name ↑	Backend type	
☐	second-backend-service	Backend service	
☐	special-topics-backend-service	Backend service	

Frontend

Protocol ↑	IP:Port	Certificate
HTTPS	34.117.148.241:443	test-cert

The Components - Cloud Armor Policies

Cloud Armor policies

[+ Create policy](#)

[Refresh](#)

[Filter](#) Enter property name or value

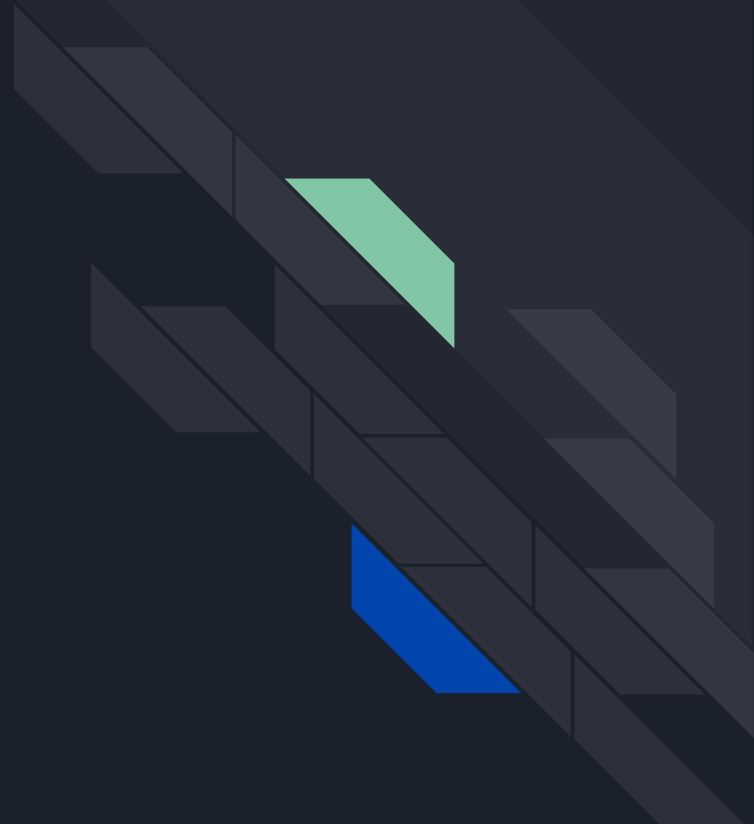
☐	Name ↑	Type
☐	default-policy	Backend security policy
☐	special-topics-waf	Backend security policy

- **Policy Type** (Scope of Policy)
- **Rule Match Conditions** (Basic or Advanced Mode)
- **Action Options** (Allow or Deny)
- **Addons**
- **Deployment** (Preview and Priority)

The Components - Preconfigured WAF Rules

☐	Action	Type	Match	Description	Priority ↑
☐	Deny (403)		evaluatePreconfiguredWaf('sqli-v33-stable')	Block SQL injection	1,000
☐	Deny (403)		evaluatePreconfiguredWaf('methodenforcement-v33-stable')	Block Method Enforcement	1,010
☐	Deny (403)		evaluatePreconfiguredWaf('xss-v33-stable')	Block Cross-Site Scripting	1,020
☐	Deny (403)		evaluatePreconfiguredWaf('scannerdetection-v33-stable')	Block Scanner Detection	1,030
☐	Deny (403)		evaluatePreconfiguredWaf('cve-canary')	Block CVE	1,040
☐	Deny (403)		evaluatePreconfiguredWaf('rce-v33-stable')	Block RCE	1,050
☐	Deny (403)	IP addresses/ranges	87.102.0.0/24	Blocking this guy in particular	1,060
☐	Deny (403)		evaluatePreconfiguredWaf('php-v33-stable')	Block PHP Injection Attack	1,070
☐	Deny (403)		evaluatePreconfiguredWaf('protocolattack-v33-stable')	Block Protocol Attack	1,080
☐	Deny (403)		evaluatePreconfiguredWaf('lfi-v33-stable')	Block Local File Inclusion	1,090
☐	Deny (403)		evaluatePreconfiguredWaf('rfi-v33-stable')	Block Remote File Inclusion	1,100
☐	Deny (403)		evaluatePreconfiguredWaf('sessionfixation-v33-stable')	Block Session Fixation	1,110
☐	Deny (403)		evaluatePreconfiguredWaf('java-v33-stable')	Block Java Attack	1,120
☐	Deny (403)		evaluatePreconfiguredWaf('nodejs-v33-stable')	Block Node.js Attack	1,130
☐	Allow	IP addresses/ranges	* (All IP addresses)	Default rule, higher priority overrides it	2,147,483,647

Let's see it in action



Sorry no Terraform

